

Assignment #1

Class:

BSCS

Subject:

DLD.

Submitted to:

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Section 2-1

Q#1 What is the weight of 7 in each of the following decimal numbers?

(a) 1947

→ The digit 7 has a weight of 1, which is 10^0 , as indicated by its position.

(b) 1799

→ 7 has a weight of 100, which is 10^2 .

(c) 1979

→ 7 has a weight of 10, which is 10^1 .

Q#2 Express each of the following decimal numbers as a power of 10.

(a) 1000

$$= 10^3$$

(c) 1000000000

$$= 10^9$$

(b) 100000000

$$= 10^7$$

Q#3:-

Give the value of each digit in the following decimal numbers.

(a) 263

Value of 2: 200

Value of 6: 60

Value of 3: 3

for sum: $200 + 60 + 3 = 263$

(b) 5436

Value of 5: 5000

Value of 4: 400

Value of 3: 30

Value of 6: 6

for sum = $5000 + 400 + 30 + 6$
= 5436

(c) 234543

Value of 2: 200000

Value of 3: 30000

Value of 4: 4000

Value of 5: 500

Value of 4: 40

Value of 3: 3

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for sum = $200000 + 30000 + 4000 + 500 + 40 + 3$
= 234543

Q#4 How high can you count with six decimal digit?

Ans:

= 999999

Section 2-2

Q#5:- Convert the following binary to decimal.

(a) 001

= $0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

= $0 + 0 + 1$

= 1

(b) 010

= $0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$

= $0 + 2 + 0$

= 2

101

$$= 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 4 + 0 + 1$$

$$= 4 + 1$$

$$= 5$$

(d) 110

$$= 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 4 + 2 + 0$$

$$= 6$$

(e) 1010

$$= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 8 + 0 + 2 + 0$$

$$= 10$$

(f) 1011

$$= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 8 + 2 + 1$$

$$= 11$$

(g) 1110

$$= 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 8 + 4 + 2 + 0$$

$$= 14$$

(h) 1111

$$= 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 8 + 4 + 2 + 1$$

$$= 15$$

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Q#6: Convert the following binary numbers into decimal.

(a) 100001

$$= 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 32 + 1$$

$$= 33$$

(b) 100111

$$= 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 32 + 0 + 0 + 4 + 2 + 1$$

$$= 39$$

(c) 101010

$$= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$

$$= 32 + 8 + 2$$

$$= 42$$

(d) 111001

$$= 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 32 + 16 + 8 + 1$$

$$= 57$$

(e) 1100000

$$= 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 64 + 32 + 0 + 0 + 0 + 0 + 0$$

$$= 96$$

Q#7: Convert each binary number to decimal.

(a) 110011.11

$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2}$$

$$= 32 + 16 + 2 + 0.5 + 0.25$$

$$= 51.75 \text{ Ans}$$

(b) 101010.01

$$= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2}$$

$$= 32 + 8 + 2 + 0.25$$

$$= 42.25 \text{ Ans}$$

(c) 100001.111

$$= 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 64 + 1 + 0.5 + 0.25 + 0.125$$

$$= 64.875 \text{ Ans}$$

Q#8 What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?

(a) two

$$\therefore 2^n - 1$$

$$= n = 2$$

(b) 10

$$\therefore 2^n - 1$$

$$n = 10$$

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$$= 2^2 - 1$$

$$= 4 - 1$$

$$= 3$$

(b) three

$$\therefore 2^n - 1, n = 3$$

$$= 2^3 - 1$$

$$= 8 - 1$$

$$= 7$$

(c) four

$$\therefore 2^n - 1$$

$$n = 4$$

$$= 2^4 - 1$$

$$= 16 - 1$$

$$= 15$$

$$= 2^{10} - 1$$

$$= 1023$$

(d) five

$$\therefore 2^n - 1, n = 5$$

$$= 2^5 - 1$$

$$= 32 - 1$$

$$= 31$$

(j) eleven

$$\therefore 2^n - 1$$

$$n = 11$$

$$= 2^{11} - 1$$

$$= 2048 - 1$$

$$= 2047$$

Q#9: How many bits are required to represent the following decimal numbers?

(a) 5

$$\therefore 2^n - 1$$

→ In it we use (n) as a value through that we get the nearest and greater or equal than our question

And n is our answer
like

$$= 2^3 - 1$$

$$= 8 - 1$$

$$= 7$$

if we get

$$2^n - 1 = 2^2 - 1$$

$$= 4 - 1$$

$$= 3$$

lowest to 5

So 3-bit is answer.

(b) 10

$$\therefore 2^n - 1$$

$$= 2^4 - 1$$

$$= 16 - 1$$

$$= 15$$

= 15-bits is answer

(c) 15

$$\therefore 2^n - 1$$

$$= 2^4 - 1$$

$$= 16 - 1$$

$$= 15$$

= 15-bits is answer

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Q#10:- Generate the
binary sequence
for each decimal
sequence.

(a) 0 through 7

Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111

(b) 8 through 15

Decimal	Binary
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

(c) 152 through 631

Decimal	Binary
16	10000
17	10001
18	10010
19	10011
20	10100
21	10101
22	10110
23	10111
24	11000
25	11001
26	11010
27	11011
28	11100
29	11101
30	11110
31	11111

Section 2-3

Q# 11:

Convert each decimal
number to binary by
using sum-of-weights

method.

(a) 12

$$\begin{aligned}
 &= 8 + 4 + 0 + 0 \\
 &= 2^3 + 2^2 + 0 + 0 \\
 &= 1100
 \end{aligned}$$

(b) 15

$$\begin{aligned}
 &= 8 + 4 + 1 \\
 &= 2^3 + 2^2 + 2^1 + 2^0 \\
 &= 1111
 \end{aligned}$$

(c) 25

$$\begin{aligned}
 &= 16 + 8 + 1 \\
 &= 16 + 8 + 1 \\
 &= 2^4 + 2^3 + 0 + 0 + 2^0 \\
 &= 11001
 \end{aligned}$$

(d) 50

$$\begin{aligned}
 &= 32 + 16 + 2 \\
 &= 2^5 + 2^4 + 0 + 0 + 2^1 + 0 \\
 &= 110010
 \end{aligned}$$

(e) 65

$$\begin{aligned}
 &= 64 + 1 \\
 &= 2^6 + 0 + 0 + 0 + 0 + 0 + 2^0 \\
 &= 2^6 + 2^0 \\
 &= 1000001
 \end{aligned}$$

Q#12:- Convert each decimal fraction to binary by using the sum of weights method.

(a) 0.26

$$= 0.25 + 0.01$$

$$= 2^{-2} + 2^{-6}$$

$$= 0.010001 \text{ Ans}$$

(b) 0.762

$$= 0.5 + 0.25 + 0.015$$

$$= 2^{-1} + 2^{-2} + 2^{-6}$$

$$= 0.110001$$

(c) 0.625

$$= 0.625$$

$$= 0.5 + 0.125$$

$$= 2^{-1} + 2^{-3}$$

$$= 0.101 \text{ Ans}$$

(d) 0.0975

$$= 0.0625 + 0.031 + 0.0039$$

$$= 2^{-4} + 2^{-5} + 2^{-8}$$

$$= 0.00011001$$

Q#13:- Convert each decimal to binary numbers using the repeated division by 2.

(a) 13

	Quotient	Remainder
$\frac{13}{2}$	$= 6$	$\rightarrow 1$
$\frac{6}{2}$	$= 3$	$\rightarrow 0$
$\frac{3}{2}$	$= 1$	$\rightarrow 1$
$\frac{1}{2}$	$= 0$	$\rightarrow 1$

1101

(b) 17

	Quotient	Remainder
$\frac{17}{2}$	$= 8$	$\rightarrow 1$
$\frac{8}{2}$	$= 4$	$\rightarrow 0$
$\frac{4}{2}$	$= 2$	$\rightarrow 0$
$\frac{2}{2}$	$= 1$	$\rightarrow 0$
$\frac{1}{2}$	$= 0$	$\rightarrow 1$

$(10001)_2$

(c) 23

Quotient Remainder

$$= \frac{23}{2} = 11 - 1$$

$$= \frac{11}{2} = 5 - 1$$

$$\frac{5}{2} = 2 - 1$$

$$\frac{2}{2} = 1 - 0$$

$$\frac{1}{2} = 0 - 1$$

$(10111)_2$

(d) 30

Quotient Remainder

$$\frac{30}{2} = 15 - 0$$

$$\frac{15}{2} = 7 - 1$$

$$\frac{7}{2} = 3 - 1$$

$$\frac{3}{2} = 1 - 1$$

$$\frac{1}{2} = 0 - 1$$

$(11110)_2$

Q#14:- Convert each decimal fraction to binary using repeated multiplication by 2.

(a) 0.76

	fractional	integers
0.76×2	1.52	1
0.52×2	1.04	1
0.04×2	0.08	0
0.08×2	0.32	0

$(0.1100)_2$

(b) 0.456

	fractional	Integers
0.456×2	0.912	0
0.912×2	1.824	1
0.824×2	1.648	1
0.648×2	1.296	1

$(0.0111)_2$

(c) 0.8732

	fractional	integers
0.8732×2	1.7464	1
0.7464×2	1.4928	1
0.4928×2	0.9856	0
0.9856×2	1.9712	1

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.9713x2	1.9424	1
.9424x2	1.8848	1

(0.110111)₂

Section 2-4

Q#15 Add the binary numbers.

(a)
$$\begin{array}{r} 10 \\ + 10 \\ \hline \end{array}$$

$$\begin{array}{r} 10 \\ + 10 \\ \hline 100 \end{array}$$

(b) $10 + 11$

$$\begin{array}{r} 10 \\ 11 \\ \hline 101 \end{array}$$

(c) $100 + 11$

$$\begin{array}{r} 100 \\ 11 \\ \hline 111 \end{array}$$

(d) $111 + 101$

$$\begin{array}{r} 111 \\ 101 \\ \hline 1100 \end{array}$$

(e) $1111 + 111$

$$\begin{array}{r} 1111 \\ 111 \\ \hline 10110 \end{array}$$

(f) $1111 + 1111$

$$\begin{array}{r} 1111 \\ + 1111 \\ \hline 11110 \end{array}$$

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Q#16 Use direct subtraction on the following binary numbers.

(a) $10 - 1$

$$\begin{array}{r} 10 \\ - 1 \\ \hline 1 \end{array}$$

(b)

$$\begin{array}{r} 100 \\ 11 \\ \hline 001 \end{array}$$

(c)

$$\begin{array}{r} 110 \\ 100 \\ \hline 010 \end{array}$$

(d) $1111 - 11$

$$\begin{array}{r} 1111 \\ - 11 \\ \hline 1100 \end{array}$$

Q#17: Perform the following binary multiplications

(a) 11×10

$$\begin{array}{r} 11 \\ \times 10 \\ \hline 00 \\ 110 \\ \hline 110 \end{array}$$

(b) 101×11

$$\begin{array}{r} 101 \\ \times 11 \\ \hline 101 \\ 1010 \\ \hline 1111 \end{array}$$

(c) 111×110

$$\begin{array}{r} 111 \\ \times 110 \\ \hline 000 \\ 1110 \\ 11100 \\ \hline 101010 \end{array}$$

Q#18: Divide the binary numbers as indicated.

(a) $110 \div 11$

$$\begin{array}{r} 10 \\ 11 \overline{) 110} \\ \underline{11} \\ 00 \end{array}$$

(b) $1010 \div 10$

$$\begin{array}{r} 101 \\ 10 \overline{) 1010} \\ \underline{10} \\ 010 \\ \underline{10} \\ 0 \end{array}$$

(c) $1111 \div 101$

$$\begin{array}{r} 11 \\ 101 \overline{) 1111} \\ \underline{101} \\ 101 \\ \underline{101} \\ 000 \end{array}$$

Section 2-5

Q#19:- What are two ways of representing zero in 1's complement form?

1: $\rightarrow 00000000$
which is +ive zero (+0)

2: $\rightarrow 11111111$, which is -ive zero (-0)

Q#20:- How is zero represented in 2's complement form?

$\rightarrow$ There is only one representation of zero which is 00000000 (+0) because if we add 1 to 11111111 , we get 00000000 (+0) which is the same as +ive zero.

Q#21 Determine the 1's complement of each binary number.

(a) 100

= 100
After taking 1's complement
= 011

(b) 111

= 111
After taking 1's C
= 000

(c) 10111011

= 10111011
After taking 1's Complement
= 01000100

(d) 10101010

= 10101010
After taking 1's complement
= 01010101

(e) 1001010

= 1001010
After taking 1's complement
= 0110101

Q#22 Determine the 2's complement of each binary number using either method.

(a) 11

= 11

After taking 1's complement

0 0

1

0 1 → 2's complement form

(b) 110

= 110

After taking 1's complement
= 001

Add (1)

= 001

1

0 1 0

→ 2's complement form

(c) 11000111

00111001 → 2's complement form

Section 2-6

Q#23 Express each in binary as an 8-bit Sign-Magnitude number.

(a) +29

2	29
2	14 - 1
2	7 - 0
2	3 - 1
	1 - 0

00011101

(b) -85

2	85
2	42 - 1
2	21 - 0
2	10 - 1
2	5 - 0
2	2 - 1
	1 - 0

S.B ← 11010101

(c) +100

$$\begin{array}{r|l} 2 & 100 \\ \hline 2 & 50-0 \\ \hline 2 & 25-0 \\ \hline 2 & 12-1 \\ \hline 2 & 6-0 \\ \hline 2 & 3-0 \\ \hline & 1-1 \end{array}$$

= 01100100

(d) -123

$$\begin{array}{r|l} 2 & 123 \\ \hline 2 & 61-1 \\ \hline 2 & 30-1 \\ \hline 2 & 15-0 \\ \hline 2 & 7-1 \\ \hline 2 & 3-1 \\ \hline & 1-1 \end{array}$$

S.B $\leftarrow$ 11111011

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Q#24:- Express each decimal number as 8-bit numbers in the 1's complement form.

(a) -34

$$\begin{array}{r|l} 2 & 34 \\ \hline 2 & 17-0 \\ \hline 2 & 8-1 \\ \hline 2 & 4-0 \\ \hline 2 & 2-0 \\ \hline & 1-0 \end{array}$$

= 00100010

After taking 1's complement
= 11011101

(b) +57

$$\begin{array}{r|l} 2 & 57 \\ \hline 2 & 28-1 \\ \hline 2 & 14-0 \\ \hline 2 & 7-0 \\ \hline 2 & 3-1 \\ \hline & 1-1 \end{array}$$

= 00111001

for +ve integers 1's same as sign-magnitude form.

Q#25: Express each decimal number as an 8-bit in the 2'sc form.

(a) +12

2	12
2	6-0
2	3-1
	1-1

0000 1110 $\rightarrow$ 2'sc form
After taking 2's complement because it is +ve integer

(b) -68

2	68
2	34-0
2	17-0
2	8-1
2	4-0
2	2-0
2	1-0

After taking 2's complement
1 000100
1 001100
1 011100

Q#26 Express each decimal number of each signed binary number in the sign magnitude form.

(a) 10011001

$$= 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 2^4 + 2^3 + 2^0$$

$$= 16 + 8 + 1$$

$$= -25$$

$\therefore$ S.B is 1 which indicate that the integer is -ive.

(b) 01110100

$$= 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 2^6 + 2^5 + 2^4 + 2^2$$

$$= 64 + 32 + 16 + 4$$

$$= +116$$

$\therefore$ S.B is 0 which indicate that the integer is +ive.

Q#27: Determine the decimal value of each signed binary number in the 1's Complement form.

(a) 10011001

$= 10011001$

if we get 1's complement of -ive signed binary number then we add 1 in result and also add (-1) MSB.

$= -2^7 + 2^4 + 2^3 + 2^0$

$= -128 + 16 + 8 + 1$

$= -103 + 1$

$= -102$

(b) 01110100

$= 2^4 + 2^6 + 2^5 + 2^4 + 2^2$
 $= 64 + 32 + 16 + 4$
 $= +116$

Q#28: Determine the decimal value of each signed binary number in the 2's Complement form.

(a) 10011001

$= 10011001$

$= -2^7 + 2^4 + 2^3 + 2^0$

$= -128 + 16 + 8 + 1$

$= -103$

(b) 01110100

$= 01110100$

$= 2^6 + 2^5 + 2^4 + 2^2$

$= 64 + 32 + 16 + 4$

$= +116$

(c) 10111111

$= 10111111$

$= -2^7 + 2^5 + 2^3 + 2^2 + 2^1 + 2^0$

$= -128 + 32 + 8 + 4 + 2 + 1$

$= -81$

Section-2-7

Q#33: Convert each of decimal numbers to binary and add using the 2's complement form.

(a) 33 and 15

$$\begin{array}{r} 2 \overline{) 33} \\ 2 \overline{) 16-1} \\ 2 \overline{) 8-0} \\ 2 \overline{) 4-0} \\ 2 \overline{) 2-0} \\ 1-0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 15} \\ 2 \overline{) 7-1} \\ 2 \overline{) 3-1} \\ 1-1 \end{array}$$

00100001

1111

$$\begin{array}{r} = 00100001 \\ + 00001111 \\ \hline 00110000 \end{array}$$

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(b) 56 and -27

$$56 = 00111000$$

$$+27 = 011011$$

$$\text{Taking 2's complement of } +27 \\ = 100101$$

$$= 111000$$

$$+ 100101$$

$$011101 \text{ Ans}$$

(c) -46 and 25

$$\begin{array}{r} 2 \overline{) 25} \\ 2 \overline{) 12-1} \\ 2 \overline{) 6-0} \\ 2 \overline{) 3-0} \\ 1-1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 46} \\ 2 \overline{) 23-0} \\ 2 \overline{) 11-1} \\ 2 \overline{) 5-1} \\ 2 \overline{) 2-1} \\ 1-0 \end{array}$$

$$+46 = 0101110$$

$$+25 = 11001$$

$$\text{Taking 2's complement of } +46 \\ = 1010010$$

$$+ 10010$$

$$011001$$

$$1101011$$

(d) -110 and -84

2 110	2 84
2 55-0	2 42-0
2 27-1	2 21-0
2 13-1	2 10-1
2 6-1	2 5-0
2 3-0	2 2-1
1-1	1-0

$$+110 = 01101110$$

$$+84 = 01010100$$

Taking 2's complement of (+110) and (+84)

$$-110 = 10010010$$

$$-84 = 10101100$$

$$= \begin{array}{r} 10010010 \\ + 10101100 \\ \hline 00111110 \end{array}$$

So, 00111110 is final answer.

Q#32: Perform each addition in the 2's complement form.

(a) $00010110 + 00110011$

$$= \begin{array}{r} 00010110 \\ + 00110011 \\ \hline 01001001 \end{array}$$

So, 01001001 is final result.

(b) $011100000 + 10101111$

$$= \begin{array}{r} 011100000 \\ + 10101111 \\ \hline \end{array}$$

Taking 2's complement of 10101111

$$= \begin{array}{r} 01010001 \\ + 01110000 \\ \hline 11000001 \end{array}$$

So, 11000001 is final result.

Q#33: Perform each addition in the 2's complement form.

(a) $10001100 + 00111001$

= Take 2's complement of 10001100

$$\begin{array}{r} 10001100 \\ - \\ 01110100 \end{array}$$

$$\begin{array}{r} 01110100 \\ + 00111001 \\ \hline 10101101 \end{array}$$

So, 10101101 is final result.

(b) $11011001 + 11100111$

Take 2's complement of 11011001

$$\begin{array}{r} 11011001 \\ - \\ 00100111 \end{array}$$

Take 2's complement of 11100111

$$\begin{array}{r} 11100111 \\ - \\ 00011001 \end{array}$$

$$\begin{array}{r} 00100111 \\ + 00011001 \\ \hline 01000000 \end{array}$$

So, 01000000 is final result.

Q#34: Perform each subtraction in the 2's complement form.

(a) $00110011 - 00010000$

minuend = 00110011

Subtrahend = 00010000

Take 2's complement of subtrahend

$$\begin{array}{r} 00110011 \\ + 11110000 \\ \hline 10010011 \end{array}$$

Discard carry

So, 00100011 is final result.

(b) 01100101 - 11101000

minuend: 01100101

Subtrahend: 11101000

Take 2's complement of subtrahend
00011000 Add in minuend

= 00011000 Add in minuses

0 1 1 0 0 1 0 1

+ 0 0 0 1 1 0 0 0

0 1 1 1 1 1 0 1 Ans

Q#35: Multiply 01101010
by 11110001 in the
2's complement form.

multiplicand = 01101010

multiplier = 11110001

Take 2's complement of
multiplies as it is (-ive)

$$= 10001111$$

Q#35 (Remaining)

0 1 1 0 1 0 1 0

X 0 0 0 0 1 1 1 1

0 1 1 0 1 0 1 0

0 1 1 0 1 0 1 0 x

0 1 1 0 1 0 1 0 x x

0 1 1 0
0 1 1 0 1 0 1 0 X X
0 0 0 0 X X X

$\begin{array}{ccccccc} & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & x & x \end{array}$

$\begin{matrix} \circ & \circ & \circ & \circ & \circ & \circ \\ \circ & \circ & \circ & \circ & \circ & \circ & X & X & X & X \\ \circ & \circ & \circ & \circ & \circ & \circ & X & X & X & X & X \end{matrix}$

00 0 0 0 0 0 X X X X X
00 0 0 0 0 0 X X X X X

Handwritten practice lines showing numbers 0 through 9 written multiple times.

check: $2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 1024 + 512 + 256 + 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 2047$

check: $2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$
 $= 1024 + 512 + 256 + 128 + 64 + 32 + 16 + 8 + 4 + 2 = 1590$

Check :-

106

15

1590

Q#36: Divide 10001000 by 00100010 in the 2's complement form.

(Divisor: 00100010

(Dividend: 10001000

∴ Divisor and dividend have different signs so the quotient will be -ive.

= 1 0 0 0 1 0 0 0

+ 1 1 0 1 1 1 1 0 Take 2's comp of divisor

Discard carry
1 0 1 1 0 0 1 1 0
+ 1 1 0 1 1 1 1 0

Discard carry
1 0 1 0 0 0 1 0 0
+ 1 1 0 1 1 1 1 0

Discard carry
1 0 0 1 0 0 0 1 0
1 1 0 1 1 1 1 0
0 0 0 0 0 0 0 0

Add 1 as add number of divisor

0 0 0 0 0 0 0 0 + 1

= 0 0 0 0 0 0 0 1

0 0 0 0 0 0 0 1 0

0 0 0 0 0 0 1 1 + 1 = 00000100 Ans

Section 2-8

Q#37: Convert each hexadecimal number to binary:

Table:

Decimal	Hexadecimal
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

$$(a) (46)_{16}$$

$$= 0100\ 0110$$

$$= (01000110)_2$$

$$(b) (54)_{16}$$

$$= (01010100)_2$$

$$(c) (B4)_{16}$$

$$= (10110100)_2$$

$$(d) (1A3)_{16}$$

$$= (000110100011)_2$$

Q#38: Convert each binary number to hexadecimal:

$$(a) 1111$$

$$= F_{16}$$

$$(b) 1011$$

$$= B_{16}$$

$$(c) 1111$$

$$= 0001\ 1111$$

$$= 1F_{16}$$

$$(d) 10101010$$

$$= (AA)_{16}$$

Q#39: Convert each hexadecimal number to decimal.

$$(a) (42)_{16}$$

$$= 4 \times 16^1 + 2 \times 16^0$$

$$= 64 + 2$$

$$= (66)_{10}$$

$$(b) (64)_{16}$$

$$= 6 \times 16^1 + 4 \times 16^0$$

$$= 96 + 4$$

$$= (100)_{10}$$

$$(c) 2B_{16}$$

$$= 2 \times 16^1 + 11 \times 16^0$$

$$= 32 + 11$$

$$= (43)_{10}$$

Q#40: Convert each decimal number to hexadecimal:

(a) 10

$$\begin{array}{r|l} 16 & 10 \\ \hline & 0-10 \end{array}$$

= A_{16}

(b) 15

$$\begin{array}{r|l} 16 & 15 \\ \hline & 0-15 \end{array}$$

= F_{16}

(c) 32

$$\begin{array}{r|l} 16 & 32 \\ \hline & 2-0 \end{array}$$

= 20_{16}

(d) 54

$$\begin{array}{r|l} 16 & 54 \\ \hline & 3-6 \end{array}$$

= 36_{16}

Q#41: Perform the following additions:

(a) $25_{16} + 33_{16}$

$$\begin{array}{r} 2 \quad 5 \\ + 3 \quad 3 \\ \hline (5 \quad 8)_{16} \end{array}$$

(b) $43_{16} + 62_{16}$

$$\begin{array}{r} 4 \quad 3 \\ + 6 \quad 2 \\ \hline (A \quad 5)_{16} \end{array}$$

(c) $A4_{16} + F5_{16}$

=	A	4		
+	F	5		
	(1	9	9)	$16 \mid 9$
				0-9
				$16 \mid 25$
				1-9
				19

(d) $FC_{16} + AE_{16}$

=	F	C		
+	A	E		
	(1	A	A)	$16 \mid 26$
				1-10
				26

Q#42: Perform the following subtraction.

(a) $60_{16} - 39_{16}$

$$\begin{array}{r} 60 \\ - 39 \\ \hline (21)_{16} \end{array}$$

(b) $A5_{16} - 98_{16}$

$$\begin{array}{r} A5 \\ - 98 \\ \hline (07)_{16} \end{array}$$

(c) $F1_{16} - A6_{16}$

$$\begin{array}{r} F1 \\ - A6 \\ \hline 45_{16} \end{array}$$

(d) $AC_{16} - 10_{16}$

$$\begin{array}{r} AC \\ - 10 \\ \hline (9C)_{16} \end{array}$$

Section 2-9

Q#43: Convert each octal number to decimal:

Table:

Decimal	binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

" " " "

(a) 14_8

$$= (14)_8$$

$$= 1 \times 8^1 + 4 \times 8^0$$

$$= 8 + 4$$

$$= (12)_{10}$$

(b) 538

$$= 5 \times 8^1 + 3 \times 8^0$$

$$= 40 + 3$$

$$= (43)_{10}$$

(c) 6358

$$= 6 \times 8^2 + 3 \times 8^1 + 5 \times 8^0$$

$$= 384 + 24 + 5$$

$$= (413)_{10}$$

(d) 77778

$$= 7 \times 8^3 + 7 \times 8^2 + 7 \times 8^1 + 7 \times 8^0$$

$$= 3584 + 448 + 56 + 7$$

$$= (4095)_{10}$$

Q#44:

Convert each decimal number to Octal by repeated division by 8.

(a) 23

$$= \frac{23}{8} \quad \begin{array}{l} \text{Quotient} \\ \text{Remainder} \end{array}$$

$$2 \quad 7$$

$$2/8 \quad - \quad 0 \quad - \quad 2$$

$$= (27)_8$$

(b) 45

$$= \frac{45}{8}$$

$$5 \quad 5$$

$$5/8 \quad - \quad 0 \quad - \quad 5$$

$$= (55)_8$$

(c) 156

$$= \frac{156}{8}$$

$$19 \quad 4$$

$$19/8 \quad - \quad 2 \quad - \quad 3$$

$$2/8 \quad - \quad 0 \quad - \quad 2$$

$$= (234)_8$$

Q#45: Convert each octal number into binary.

(a) 178

$$= 178$$

$$= 1 \quad 7$$

$$= 001 \quad 111$$

$$= (001111)_2 \quad \text{Ans}$$

(b) 7778

= 7778

= 7 7 7

= 111 111 111

= (111111111)₂ Ans

(c) 4568

= 4568

= 4 5 6

= 100 101 110

= (100101110)₂ Ans

Q#46: Convert each binary numbers to octal:

(a) 100

= 100

= 4 Ans

(c) 1100

= 001 100

= 14 Ans

(e) 101010

= 101 010

= 52

Ans

(b) 110

= 110

= 6 Ans

(d) 1111

= 001 111

= 17 Ans

(f) 110011

= 110 011

= 63

Ans

Section 2-10

Q#47: Convert each of the following decimal numbers to 8421 BCD.

(a) 10

= 0001 + 0000

= 0001 0000 Ans

(b) 13

= 1 3

= 0001 0011

= 0001 0011 Ans

(c) 156

= 1 5 6

= 0001 0101 0110

= 0001 0101 0110 Ans

(d) 125

= 1 2 5

= 0001 0010 0101

= 0001 0010 0101 Ans

Q#49: Convert the following decimal numbers to BCD.

(a) 1051

= 1 0 5 1
= 0001 0000 0101 0001
= 0001 0000 0101 0001 Ans

(b) 547

= 5 4 7
= 0101 0100 0111
= 0101 0100 0111 Ans

(d) 210

= 2 1 0
= 0010 0001 0000
= 0010 0001 0000 Ans

Q#50: Convert each of the decimal of the following BCD.

(a) 0001

= 0001
= 1

(b) 0110

= 0110
= 6

(c) 00110010

= 0011 0010
= 32

(d) 10011000

= 1001 1000
= 9 8
= 98

Q#51: Convert each of the BCD numbers to decimal.

(a) 1000 0000

= 1000 0000
= 80

(b) 0100 0101 0001

= 0100 0101 0001
= 47 2 1
= 421

(c) 0111 0101 0100

= 0111 0101 0100
= 7 5 4
= 754

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Q#52: Add the following BCD numbers.

(a) $1000 + 0001$

$$\begin{array}{r} 1000 \\ + 0001 \\ \hline 1001 \end{array} \quad \begin{array}{r} 8 \\ + 1 \\ \hline 9 \end{array}$$

Ans 9

(b) $0111 + 0010$

$$\begin{array}{r} 0111 \\ + 0010 \\ \hline 1001 \end{array} \quad \begin{array}{r} 7 \\ + 2 \\ \hline 9 \end{array}$$

Ans 9

(c) $0001000 + 0001000$

$$\begin{array}{r} 0001000 \\ + 0001000 \\ \hline 0010000 \end{array} \quad \begin{array}{r} 18 \\ + 11 \\ \hline 29 \end{array}$$

Ans 29

(d) $01100100 + 00110011$

$$\begin{array}{r} 01100100 \\ + 00110011 \\ \hline 10010111 \end{array} \quad \begin{array}{r} 64 \\ + 33 \\ \hline 97 \end{array}$$

Ans 97

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Q#53: Add the following BCD numbers.

(a) $1000 + 0110$

$$\begin{array}{r} 1000 \\ + 0110 \\ \hline 1110 \end{array} \rightarrow > 9$$

110100

Ans 10100

(b) $00010101 + 00100111$

$$\begin{array}{r} 00100101 \\ + 00100111 \\ \hline 01001100 \end{array} \quad \begin{array}{r} 25 \\ + 27 \\ \hline 52 \end{array}$$

Ans 52

(c) $1001 + 0111$

$$\begin{array}{r} 1001 \\ + 0111 \\ \hline 10000 \end{array} \quad \begin{array}{r} 9 \\ + 7 \\ \hline 16 \end{array}$$

Ans 16

Q#54: Convert each pair of decimal numbers to BCD and add as indicated.

(a) $4 + 3$

$$= 0000 + 0011$$

$$\begin{array}{r} 0100 \\ + 0011 \\ \hline 0111 \end{array} \quad \begin{array}{r} 4 \\ 3 \\ \hline 7 \end{array} \quad \text{Ans}$$

(b) $5 + 2$

$$= 0101 + 0010$$

$$\begin{array}{r} 0101 \\ + 0010 \\ \hline 0111 \end{array} \quad \begin{array}{r} 5 \\ 2 \\ \hline 7 \end{array} \quad \text{Ans}$$

(c) $113 + 101$

$$\begin{array}{r} 00010001001 \\ + 00010000001 \\ \hline 00010001001 \\ + 00010000001 \\ \hline 001000010100 \end{array} \quad \begin{array}{r} 113 \\ 101 \\ \hline 214 \end{array} \quad \text{Ans}$$

Section 2-11

Q#56: Convert each binary number to Gray code.

(a) 11011

$$\begin{array}{ccccccc} & 1 & + & 1 & + & 0 & + & 1 & + & 1 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ & 1 & & 0 & & 1 & & 1 & & 0 \end{array}$$

$$= (10110) \text{ (Gray code)}$$

(b) 1001010

$$\begin{array}{ccccccc} & 1 & + & 0 & + & 0 & + & 1 & + & 0 & + & 1 & + & 0 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ & 1 & & 0 & & 1 & & 1 & & 1 & & 1 & & 0 \end{array}$$

$$= (110111) \text{ (Gray code)}$$

(c) 1111011101110

$$\begin{array}{ccccccccccccccc} & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & \end{array}$$

$$= (1000110011001) \text{ (Gray code)}$$

Q#57: Convert each Gray code to BCD.

(a) 1010

= 1 0 1 0
 $\downarrow \nearrow \downarrow \nearrow \downarrow \nearrow \downarrow \nearrow$
 = 1 1 0 0
 = (1100) BCD

(b) 00010

= 0 0 0 1 0
 $\downarrow \nearrow \downarrow \nearrow \downarrow \nearrow \downarrow \nearrow$
 = 0 0 0 1 1
 = (00011) BCD

(c) 1100001001

= 1 1 0 0 0 0 1 0 0 1
 $\downarrow \nearrow \downarrow \nearrow \downarrow \nearrow \downarrow \nearrow \downarrow \nearrow \downarrow \nearrow \downarrow \nearrow$
 = 1 0 0 0 0 0 1 1 1 0
 = (1000001110) BCD

(Section 2-12)

Q#63: Determine which of the following even of parity codes are in error.

(a) 10010010

(b) 01101010

(c) 101111001010

→ In 01101010 even parity codes in error.

Q#64: Determine which of the following odd of parity codes are in error.

(a) 11110110

(b) 00100101

(c) 010101010101

→ In 11110110 odd parity codes in error.

→ In 010101010101 odd parity codes in error.

Q#65: Attach the proper even parity bit to each of the following bytes of data.

(a) 10100100

we add 1 parity to the left-most side

= 110100100

↓ Is parity bit

(b) 00001001

→ we add 0 parity bit to the left-most side.

= 00001001

↓ Is parity bit.

(c) 11111110

→ we add 1 parity bit to the left-most side

= 111111110

↓ Is parity bit

Q#68: Apply CRC to the data bits 10110010 using the generator code 1010 to produce the transmitted CRC code.

Data: 10110010

Generator: 1010

→ Since the generator code has four bits, add four 0s to the data.

The appended data is

D' = 101100100000

→ Divide the appended data by using modulo-2 operation until all bits have been used.

D' = 101100100000
G = 1010

[illegible]

code: 001100100110

$G = 1010$

→ Add 4 0's on right side of data bit since generator has four digits.

$$\begin{array}{r} 0011001001100000 \\ 1010 \overline{) } \\ \underline{1001 } \\ 1100 \\ 1010 \overline{) } \\ \underline{1101 } \\ 1010 \overline{) } \\ \underline{1110 } \\ 1010 \overline{) } \\ \underline{1000 } \\ 1010 \overline{) } \\ \underline{1011 } \\ 1010 \overline{) } \\ \underline{1000 } \\ 1010 \overline{) } \\ \underline{1000 } \\ 1010 \overline{) } \\ \underline{0010 } \end{array}$$

Again using appended the data remainder

$D' = 0011001001100000$

Again divided by generator using modulo-2 operation.

$D' = 0011001001100000$

$G = 1010$

$$\begin{array}{r} 0011001001100000 \\ 1010 \overline{) } \\ \underline{1001 } \\ 1100 \\ 1010 \overline{) } \\ \underline{1101 } \\ 1010 \overline{) } \\ \underline{1110 } \\ 1010 \overline{) } \\ \underline{1000 } \\ 1010 \overline{) } \\ \underline{1011 } \\ 1010 \overline{) } \\ \underline{1000 } \\ 1010 \overline{) } \\ \underline{1010 } \\ 0000 \end{array}$$

∴ Remainder = 0

It is correct code.

Q#66: Apply modulo-2 to the following:

According to modulo-2 Operation:

Input		Output
0	0	0
0	1	1
1	0	1
1	1	0

Also called exclusive OR.

(a) $1100 + 1011$

$= 1100 + 1011$

Applying modulo-2

$= \begin{array}{r} 1100 \\ + 1011 \\ \hline \end{array}$

$\begin{array}{r} 1100 \\ + 1011 \\ \hline \end{array}$

$\begin{array}{r} 1100 \\ + 1011 \\ \hline 0111 \end{array} \rightarrow \text{final result}$

(b) $1111 + 0100$

$= 1111 + 0100$

Applying modulo-2-operation

$= \begin{array}{r} 1111 \\ + 0100 \\ \hline \end{array}$

$\begin{array}{r} 1111 \\ + 0100 \\ \hline \end{array}$

$\begin{array}{r} 1111 \\ + 0100 \\ \hline 1011 \end{array} \rightarrow \text{final result}$

$$(c) \ 10011001 + 100011100$$

$$= 10011001 \quad 100011100$$

Applying modulo-2 operation

$$= \begin{array}{ccccccc} 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}$$

$$+ \begin{array}{ccccccc} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \end{array}$$

$$\begin{array}{ccccccc} 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{array}$$
